

[5870] - 1113

T.E. (Chemical)

CHEMICAL ENGINEERING THERMODYNAMICS**(2019 Pattern) (Semester - I) (309344)****Time : 2½ Hours]****[Max. Marks : 70****Instructions to the candidates :**

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to right side indicates full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) Assuming the validity of Raoult's law do the following calculations for benzene(1) toluene (2) system. **[18]**

- a) Given $x_1 = 0.33$ and $T = 100^\circ\text{C}$, find y_1 and P
- b) Given $y_1 = 0.33$ and $T = 100^\circ\text{C}$, find x_1 and P
- c) Given $x_1 = 0.33$ and $P = 120$ KPa, find y_1 and T .

The Antoine equation and parameters are given below:

$$\ln P^{\text{sat}}/\text{KPa} = A - \frac{B}{T+C} \quad (T \text{ in } ^\circ\text{C})$$

Compound	A	B	C
Benzene	13.8594	2773.78	220.07
Toluene	14.0098	3103.01	219.79

OR

- Q2) a)** Construct the p - x - y diagram for cyclohexane (1) – benzene (2) system at 313K given that at 313 K the vapour pressures of cyclohexane and benzene are 24.62 KPa and 24.41 KPa respectively. The liquid phase activity coefficients are given by $\ln \gamma_1 = 0.458 x_2^2$, $\ln \gamma_2 = 0.458 x_1^2$. **[10]**
- b)** Explain phase rule for non- reacting systems with suitable examples. **[8]**

P.T.O.

- Q3) a)** Explain liquid - liquid equilibrium diagram on triangular co-ordinates for a system in which two pairs are partially soluble. [8]
- b) Explain the following two methods of consistency tests for VLE data:[9]
- Using the coexistence equation.
 - Using the partial pressure data.

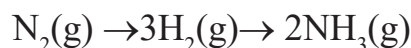
OR

- Q4) a)** Derive the following expression for solid - liquid equilibrium. [10]

$$\Psi_i = \exp \int_{T_m}^T \frac{H_i^l - H_i^s}{RT^2} dt \quad \text{Where } \Psi_i = \frac{f_i^s}{f_i^l} \text{ and } H_i^s \text{ and } H_i^l \text{ are the molar enthalpies for the species } i \text{ in the liquid and solid phases. } T_m \text{ is the melting temperature of pure species } i.$$

- b) Discuss various criteria for phase equilibrium. [7]

- Q5) a)** Estimate the standard free energy change and equilibrium constant at 700K for the reaction [14]



Given that the standard heat of formation and standard free energy of formation of ammonia at 298K to be- 46100 J/mol and - 16500 J/mol respectively. The specific heat data is given as,

$$C_p = 27.27 + 4.93 \times 10^{-3}T \text{ for } N_2$$

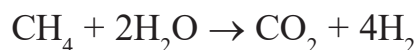
$$C_p = 27.01 + 3.51 \times 10^{-3}T \text{ for } H_2$$

$$C_p = 29.75 + 25.11 \times 10^{-3}T \text{ for } NH_3$$

- b) Write a note on feasibility of a chemical reaction. [4]

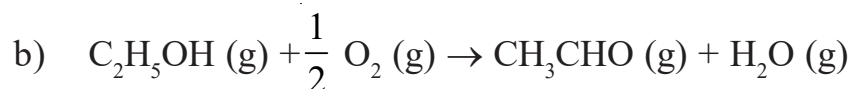
OR

- Q6) a)** A system consisting 2 mol methane and 3 mol water is undergoing the following reactions: $CH_4 + H_2O \rightarrow CO + 3H_2$ [9]



Derive expressions for mole fractions in terms of extent of reaction.

- b) Calculate the standard Gibbs free energy change and the equilibrium constant at 298.15K for the following reactions: [9]



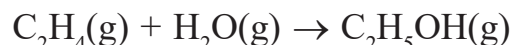
Free energies of formation at 298.15 K are given as follows:

$$\Delta G_f^\circ(\text{NH}_3) = -16.747 \text{ kJ/mol}, \Delta G_f^\circ(\text{CH}_3\text{CHO}) = -133.978 \text{ kJ/mol}$$

$$\Delta G_f^\circ(\text{H}_2\text{O}) = -228.600 \text{ kJ/mol}, \Delta G_f^\circ(\text{C}_2\text{H}_5\text{OH}) = -174.883 \text{ kJ/mol}.$$

- Q7) a) 100 moles of gas mixture containing 60% H_2 , 20% N_2 and 20% inerts is catalytically reacted to get NH_3 at 50 bar and 400°C. Calculate percent conversion of H_2 and percent NH_3 in exit gases. Given K_p at 400°C = 0.0125. [8]

- b) Calculate the equilibrium constant at 298 K for the reaction [9]



with the help of following data. Also comment about the feasibility of reaction.

	$S_{298}^\circ, \text{J/mol.K}$	$H_{298}^\circ, \text{J/mol.K}$
$\text{C}_2\text{H}_4(\text{g})$	220.85	48,986
$\text{H}_2\text{O}(\text{g})$	189.12	-241997
$\text{C}_2\text{H}_5\text{OH}(\text{g})$	278.00	-238941

OR

- Q8) a) A feed stock of pure n- butane is cracked at 750K and 1.2 bar to produce olefins. Only two reactions have favourable equilibrium conversion at these condition: [10]



if these reactions reach equilibrium, what is the product composition?

- b) Explain in detail about the relation of equilibrium constant to composition. [7]



[5926]-26

T.E. (Chemical)

CHEMICAL ENGINEERING THERMODYNAMICS

(2019 Pattern) (Semester - I) (309344)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary.

Q1) a) A liquid mixture of cyclohexane (1) /phenol (2) for which $x_1 = 0.6$ is in equilibrium with its vapour at 417 K. Determine the equilibrium pressure P and vapour composition y_1 from the following information. **[10]**

$$\ln \gamma_1 = A x_2^2, \ln \gamma_2 = A x_1^2, \text{ At } 417\text{K } P_1^{(\text{sat})} = 75.20 \text{ KPa}, P_2^{(\text{sat})} = 31.66 \text{ KPa}.$$

The system forms an azeotrope at 417K for which $x_1^{\text{az}} = y_1^{\text{az}} = 0.294$.

b) Explain the Gibbs - Duhem equation and its various forms. **[8]**

OR

Q2) Assuming the validity of Raoult's law do the following calculations for benzene (1)/toluene (2) system. **[18]**

- a) Given $x_1 = 0.33$ and $T = 100^\circ\text{C}$, find y_1 and P
- b) Given $y_1 = 0.33$ and $T = 100^\circ\text{C}$, find x_1 and P
- c) Given $x_1 = 0.33$ and $P = 120 \text{ KPa}$, find y_1 and T.

The Antoine equation and parameters are given below:

$$\ln P^{\text{sat}}/\text{KPa} = A - \frac{B}{T+C} \quad (T \text{ in } ^\circ\text{C})$$

Compound	A	B	C
Benzene	13.8594	2773.78	220.07
Toluene	14.0098	3103.01	219.79

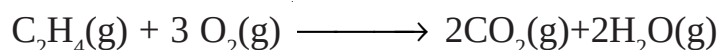
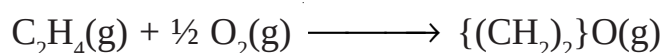
P.T.O.

- Q3) a)** Explain following consistency tests for VLE data: **[10]**
- i) Slope of $\ln \gamma_1$ curves
 - ii) Redlich Kister method.
- b) Explain triple point and eutectic temperature with neat diagram. **[7]**

OR

- Q4) a)** Explain liquid - liquid equilibrium diagram on triangular co-ordinates for a system in which two pairs are partially soluble. **[8]**
- b) Explain the following two methods of consistency tests for VLE data: **[9]**
- i) Using the coexistence equation.
 - ii) Using the partial pressure data.

- Q5) a)** A system initially containing 2 mol C_2H_4 and 3 mol O_2 undergoes the reactions: **[10]**

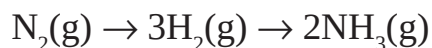


Develop expressions for the mole fractions of the reacting species as functions of the reaction co-ordinates for the two reactions.

- b) Derive the following expression. **[8]**
- $$-RT \ln K = \sum v_i G_i^0$$

OR

- Q6) a)** Estimate the standard free energy change and equilibrium constant at 700 K for the reaction **[14]**



Given that the standard heat of formation and standard free energy of formation of ammonia at 298 K to be -46100 J/mol and -16500 J/mol respectively. The specific heat data is given as,

$$C_p = 27.27 + 4.93 \times 10^{-3} T \text{ for } N_2$$

$$C_p = 27.01 + 3.51 \times 10^{-3} T \text{ for } H_2$$

$$C_p = 29.75 + 25.11 \times 10^{-3} T \text{ for } NH_3.$$

- b) Write a note on feasibility of a chemical reaction. **[4]**

Q7) a) For the cracking reaction $\text{C}_3\text{H}_8 (\text{g}) \longrightarrow \text{C}_2\text{H}_4 (\text{g}) + \text{CH}_4 (\text{g})$ the equilibrium conversion is negligible at 300 K but becomes appreciable at temperatures above 500K. For a pressure of 1 bar, determine **[12]**

- i) Fractional conversion at 600K
- ii) The temperature at which the fractional conversion is 80%.

b) Explain phase rule for reacting systems. **[5]**

OR

Q8) a) 100 moles of gas mixture containing 60% H_2 , 20% N_2 and 20% inerts is catalytically reacted to get NH_3 at 50 bar and 400°C . Calculate percent conversion of H_2 and percent NH_3 in exit gases. Given K_p at $400^\circ\text{C} = 0.0125$. **[8]**

b) Calculate the equilibrium constant at 298 K for the reaction **[9]**



with the help of following data. Also comment about the feasibility of reaction.

	$S^\circ_{298}, \text{ J/mol.K}$	$H^\circ_{298}, \text{ J/mol.K}$
$\text{C}_2\text{H}_4 (\text{g})$	220.85	48,986
$\text{H}_2\text{O} (\text{g})$	189.12	-241997
$\text{C}_2\text{H}_5\text{OH} (\text{g})$	278.00	-238941

x x x